The United Illuminating Company
100 Marsh Hill Road
Orange, CT 06477

DISTRIBUTION PLANNING
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07/28/2026

VA Hospital
1 West Spring St, West Haven
New Customer Load Report



VA Hospital, West Haven New Customer Load Report

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Background

The US Department of Veterans Affairs West Haven Medical Center ("VA Hospital") is adding a new surgical tower and will be upgrading boiler and chiller plant facilities at 1 West Spring Street, West Haven adding additional commercial space to this property. The customer is requesting to add this new load to the UI distribution system through the same 13.8 kV distribution feeders that are currently supplying the VA Hospital facilities. These feeders are Elmwest Substation Feeders 626 and 627. In addition, the customer is requesting load study considering campus electrical service alterations so that one UI circuit is from a different substation, relocation of one of the existing service feeders off campus with 50 feet of separation between feeders while maintaining the same Vista switch terminations, and to consider burying feeder 626 along West Spring Street so it is underground for its entire run and is not co-located on a utility pole with the alternate hospital supply. This project will be located on West Spring Street between Stevens Avenue and Campbell as shown in Figure 1.

The two new customer construction projects will be connected to the existing utility service at CHP plant. One is for the new surgical tower approximately 2.3 MW connected load and the other is for new boiler and chiller plants with 1.2 MW connected load. The customer's existing demand is assumed to be 5 MW, and the highest peak during past three years was around 4.4 MW (189 Amp) since temporary generation was operating. The total new demand is 8.5 MW, which after applying a diversifying factor of 90% brings about a new total load around 7.92 MW (327 Amp) for capacity planning purposes by UI.

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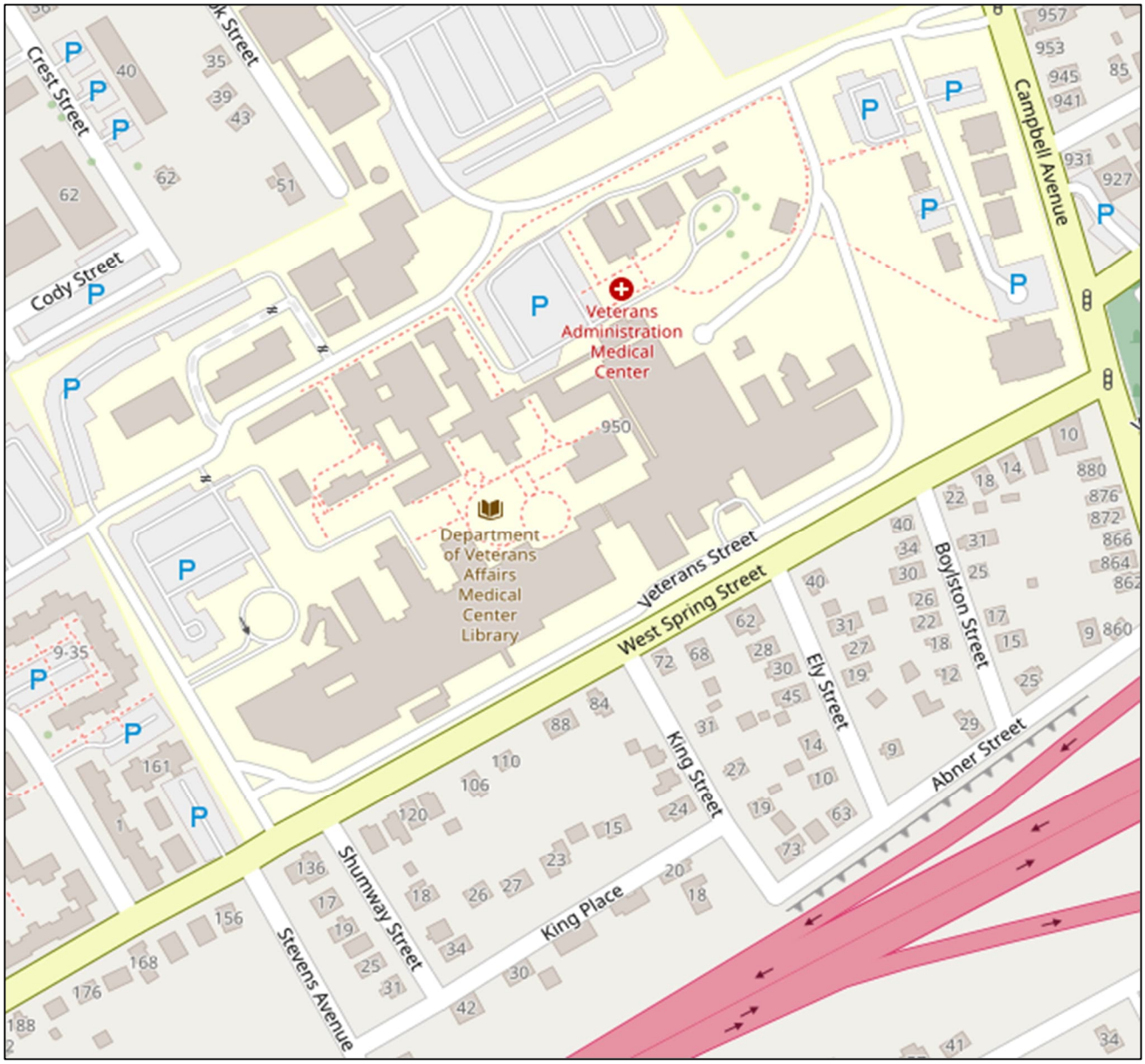


Figure 1: Customer Location

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Feeder Loading Analysis – Existing Conditions (See Appendix 3 for One-line diagrams)

First UI inspected if it would be possible to supply the new load via circuits 626 and 627 of Elmwest Substation, which are also supplying the existing load. In this case, circuit 626 will be the normal supply and 627 will be the backup supply. Both circuits 626 and 627 operate normally closed at the Vista switch (under UI's control), followed by the customer's switchgear with the incoming circuit breaker 52-L1 normally closed for Circuit 626 and circuit breaker 52-L2 normally opened for Circuit 627 and customer bus tie normally closed (See appendix 3 one-line). For the loss of service to either one of the circuits, the remaining circuit will continue to be in service.

The previous three-year peak load on circuit 626 was 267 Amp at 13.8 kV and the normal rating for this circuit is 391 Amps at 13.8 kV. The table below summarizes the peak load, rating, and remaining capacity for this feeder. The available capacity is calculated after applying a 90% threshold to the normal rating.

Table 1: Existing conditions of Circuit 626

			Loading (Amps)	
Feeder	**Normal**	**Contingency**	**Peak Load**	**Normal Available Capacity (Amps) ***
626	391	521	267	85

***** Per Avangrid Distribution Planning Criteria, the thermal loading of the wire and cable portions of a circuit should not exceed the ratings as documented in UI's Distribution Engineering Guide (DEG) 10.3 under normal conditions.

Table 2 summarizes the peak load, rating and remaining contingency capacity for circuits 623, 625 and 627, which are the backups to circuit 626. If circuit 626 faults, the VA Hospital (327 Amps) is automatically thrown over to Circuit 627 at hospital switchgear while the Campbell Avenue overhead load (83 Amps) is manually shifted to either circuit 623 or 625.

Table 2: Available emergency capacity on Circuits 623, 625, and 627

	Rating (Amps)		Loading (Amps)		
Feeder	**Normal**	**Contingency**	**Peak Load**	**Queued Load**	**Available Contingency Capacity**
623	391	521	108	5	408
625	487	599	113	0	486
627	487	599	370	5	224

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Evaluating various options (See Appendix 3 for one-line diagrams)

Circuit 626 loading table shows that the circuit cannot accommodate an additional 138 Amps of new load at 13.8 kV before the feeder reaches its rating. After the new customer load addition, circuit 626 will have a total projected load of 396 Amps at 13.8 kV with a capacity deficiency of 36 Amps.

Table 3: Available normal capacity on Circuits 626 for the new load

	Rating (Amps)		Loading (Amps)			
Feeder	**Normal**	**Contingency**	**Peak Load**	**VA Hospital New Load**	**Total Projected Load**	**Projected Available Capacity***
626	391	521	267	138	405	-53

***** This ignores UI's Distribution Planning practice to maintain a circuit's peak load at or less than 90% of its normal rating to allow for background load growth.

Table 4 summarizes the peak load, rating and remaining contingency capacity for circuits 623, 625 and 627, which are the backups to circuit 626. If circuit 626 faults, the VA Hospital (327 Amps) is automatically thrown over to circuit 627 at hospital switchgear while the Campbell Avenue overhead load (83 Amps) is manually shifted to either circuit 623 or 625. The 138 Amp new load increase at hospital results in circuit 627 having a 103 Amp contingency capacity deficiency.

Table 4: Available emergency capacity on Circuits 623, 625, and 627

	Rating (Amps)		Loading (Amps)		
Feeder	**Normal**	**Contingency**	**Peak Load**	**Queued Load**	**Projected Available Contingency Capacity**
623	391	521	108	5	408
625	487	599	113	0	486
627	487	599	370	5	224

The following options have been developed to increase available capacity of circuit 626 so there is enough capacity to supply the proposed load increase of 138 Amps as well as increase available capacity on the hospital backup circuit for contingency coverage. The customer has also requested additional options be included to help the hospital in meeting or exceeding specific resiliency

requirements regarding circuit separation and having separate substation supplies. As stipulated in the conclusion section, any option other than 4B must be further studied by Distribution Planning should the customer opt out of Option 4B.

Option 1

It is recommended to shift circuit 626 Campbell Avenue overhead distribution load (83 Amps) to circuit 625, thus making the normal feeder to hospital express. It is recommended to reconfigure circuits so that the hospital backup source is from circuit 623 overhead on West Spring Road by installing one span of 3-397 AL wire on West Spring Street between pole 5931 and pole 5932 extending circuit 623 and installing riser air-break switch and terminators on pole 5932 replacing 627 source with 623 for the hospital and eliminate circuit 627 aerial cable and aerial Y-splice. In this option, circuits 626 and 623 will be from two different buses at Elmwest Substation. The overhead portion of circuit 626 can be shifted to circuit 625 by closing air-break switch on pole 5302 Dana Street and opening riser air-break switch on pole 4448 on Campbell Ave. Circuit 626 riser air-break switch and underground cable back to SC 1068 (Y-splice) should be removed to eliminate unnecessary mainline exposure. However, this option will not meet the requirement of having the back-up feeder from a different substation, and therefore, is abandoned.

Option 2

In this option, it is recommended to shift circuit 626 Campbell Avenue overhead distribution load (83 Amps) to circuit 623 and thus making the normal feeder to hospital express. It is recommended to reconfigure circuits so that the hospital backup source is from circuit 625 by breaking down circuit 627 aerial Y-splice and making up aerial Y-splice on Campbell Avenue with circuit 625, thus replacing backup to the hospital. It would also be necessary to swap feeder 625 and 632 feeder positions at Elmwest Substation to have both hospital normal and backup sources on separate buses. The overhead portion of circuit 626 can be shifted by closing air-break switch on pole 1727 Campbell Ave and opening riser air-break switch on pole 4448 Campbell Ave. Circuit 626 riser air-break switch and underground cable back to SC 1068 (Y-splice) should be removed to eliminate unnecessary mainline exposure. However, similar to Option 1, this option will not meet the requirement of having the back-up feeder from a different substation, and therefore, is abandoned.

Option 3

Like option 2 it is recommended to shift circuit 626 Campbell Avenue overhead distribution load (83 Amps) to circuit 623 thus making the normal feeder to the hospital an express feed. It is recommended to reconfigure circuits so that the hospital backup source is from circuit 632 by installing 5,300 ft. of 500

AL preassembled aerial cable on West Spring Street between Meloy Road pole 2597 (Y-splicing new cable with Circuit 632) and pole 5931 West Spring Street and tying into circuit 627 aerial cable to the hospital. This option provides fifty-foot aerial clearance between hospital feeders although these feeders would still share common underground duct-line just outside Elmwest substation. Circuit 626 riser air-break switch and underground cable back to SC 1068 (Y-splice) should be removed to eliminate unnecessary mainline exposure. However, the numerical results for this option are skipped because after considering pole guying/loading and existing field conditions it was determined by Distribution Engineering that this option is not feasible to construct.

Option 3B

Like option 2 it is recommended to shift circuit 626 Campbell Avenue overhead distribution load (83 Amps) to circuit 623 thus making the normal feeder to hospital express. It is recommended to reconfigure circuits so that the hospital backup source is from circuit 632 by installing air-break switch on Meloy Road south of West Spring Street on a new pole between P2597 and P1176 and closing tie recloser pole 5508 on Meloy Road and opening newly installed air-break switch on Meloy Road transferring (15 Amps) from circuit 633 to circuit 632. Then by installing air-break switch on West Spring Street just west of Campbell Ave on pole 5968 and closing tie air-break switch on West Spring Street pole 2638 and opening new air-break pole 5968 transferring 98 Amps of circuit 623 to circuit 632. It is then recommended to extend circuit 632 overhead on West Spring Road by installing one span of 3-397 AL wire on West Spring Street between pole 5931 and pole 5932 and installing riser air-break switch and terminators on pole 5932 replacing 627 source with 632 for the hospital and eliminate circuit 627 aerial cable and aerial Y-splice. This option provides greater than fifty-foot aerial clearance between hospital feeders, but these feeders would still share common underground duct-line just outside Elmwest substation. Hospital feeders would be on alternate buses for future bus splitting project. Circuit 626 riser air-break switch and underground cable back to SC 1068 (Y-splice) should be removed to eliminate unnecessary mainline exposure. This option will be used as a constituent to Option 4B, which will be discussed later on.

Option 4

Like option 2 it is recommended to shift circuit 626 Campbell Avenue overhead distribution load (83 Amps) to circuit 623 thus making the normal feeder to the hospital express. It is then recommended to reconfigure circuits so that the hospital backup source is from Allings Crossing Circuit 1433. Like option 3 by installing 5,300 ft. of 500 AL preassembled aerial cable on West Spring Street between Meloy Road near pole 2597 (Y-splicing new cable with Circuit 1433) and pole 5931 West Spring Street and tying into Circuit 627 aerial cable to the hospital.

It is then necessary to cut and swap circuit 632 and circuit 1433 aerial cables on Meloy Road c/o Sawmill Road pole 5748 or between West Spring Street and pole 5045 (existing circuit 632 riser) to facilitate hospital having alternate substation utility source. This option also provides greater than fifty-foot aerial and underground clearances between hospital feeders. Circuit 626 riser air-break switch and underground cable back to SC 1068 (Y-splice) should be removed to eliminate unnecessary mainline exposure. However, the numerical results for this option are skipped since this option is a slight variation of option 3, which was determined by Distribution Engineering to be infeasible to construct.

Option 4B

Like option 2 it is recommended to shift circuit 626 Campbell Avenue overhead distribution load (83 Amps) to circuit 623 thus making the normal feeder to the hospital express. It is then recommended to reconfigure circuits so that the hospital backup source is from circuit 1433 off Allings Crossing substation. Option 4B recommends implementing option 3B then cutting and swapping circuit 632 and circuit 1433 aerial cables on Meloy Road c/o Sawmill Road pole 5748 or between West Spring Street and pole 5045 (existing circuit 632 riser) to facilitate hospital having alternate substation utility source. This option also provides greater than fifty-foot underground clearance between hospital feeders but leaves 275 ft of aerial exposure that can be eliminated by completing Option 5. Circuit 626 riser air-break switch and underground cable back to SC 1068 (Y-splice) should be removed to eliminate unnecessary mainline exposure. Due to the adverse thermal impact of the new load on neighboring underground cables, the thermal ratings of affected circuits must be re-calculated, which have been shown in Table 5 for the main circuit (Circuit 626) and Circuit 632. As seen from the table, there will be significant capacity deficiency on Circuit 632 once VA Hospital’s new load is in service and hospital alternate source is swapped.

Table 5: Before and after conditions for Circuits 626 and 632

	Before New Load Added			After New Load Added					
Circuit	Normal Rating (Amps)	Emergency Rating (Amps)	Peak Load (A)	Normal Rating (Amps)	Emergency Rating (Amps)	New Peak Load (A)	Queued Load (A)	New Peak Load Plus Queued (A)	Remaining Normal Capacity (A)
626	391	521	267	364	521	327	0	327	1
632	391	521	68	329	498	305	84	385	-89

To mitigate the aforesaid capacity deficiency, the three underground segments from SC68 to SC71 on Circuit 632, as shown in Figure 2, must be upgraded from 500 PL/EPR to 750 EPR (approximately 850 ft). Having done so, the conditions for Circuit 632 will change as shown in Table 6. It is noted that this cable upgrade will not improve the ratings of any circuits other than Circuit 632 itself.

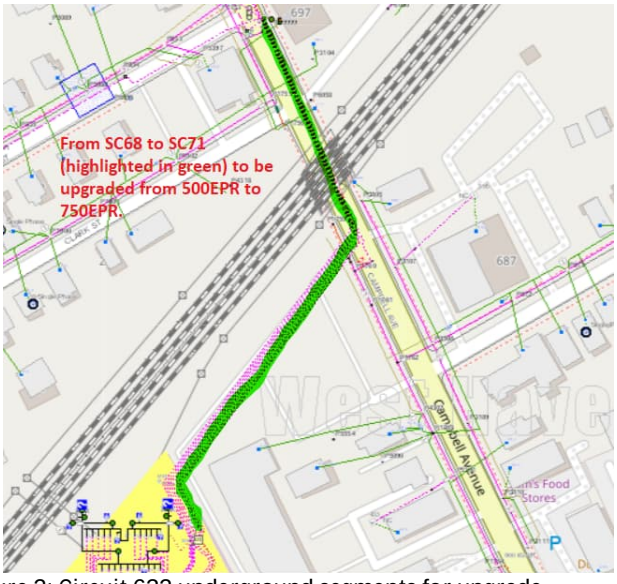


Figure 2: Circuit 632 underground segments for upgrade

Table 6: Conditions for Circuits 632 before and after the cable upgrade

	After New Load Added & Before Upgrade to 750 EPR		After New Load Added & After Upgrade to 750 EPR			
Circuit	Normal Rating (Amps)	Emergency Rating (Amps)	Normal Rating (Amps)	Emergency Rating (Amps)	Peak Load (A)	Remaining Normal Capacity (A)
632	329	498	429	599	385	1

During contingency conditions (when circuit 626 fails), the VA Hospital (327 Amps) would automatically be thrown over to Circuit 1433 at the hospital switchgear and the addition of 138 Amps of new customer load will not have any adverse effects on contingency scenario as shown in table below.

Table 7: Available emergency capacity on Circuit 1433 after Option 4B implemented

	Rating (Amps)		Loading (Amps)			
Feeder	Normal	Contingency	Peak Load (Before Option 4B)	Peak Load (Option 4B Implemented)	Queued Load	Available Contingency Capacity (Amps)
1433	429	599	305	189	0	410

Voltage analysis in CYME has revealed that swapping circuit 632 and circuit 1433 aerial cables would cause the voltage of customers along Beechwood Rd, Indian Hill Rd, Tall Timber Rd, and N Lakeview Dr to drift out of allowed range. However, the voltage issue is addressed if some of those customers are transferred from phase C to phase B. Therefore, as part of Option 4B, it is needed to perform load balancing at pole #1382 (located on Derby Turnpike) by moving the single-phase network downstream of that pole (#1382) from phase C to phase B (Road Phase to Middle Phase), as indicated in Figure 3.

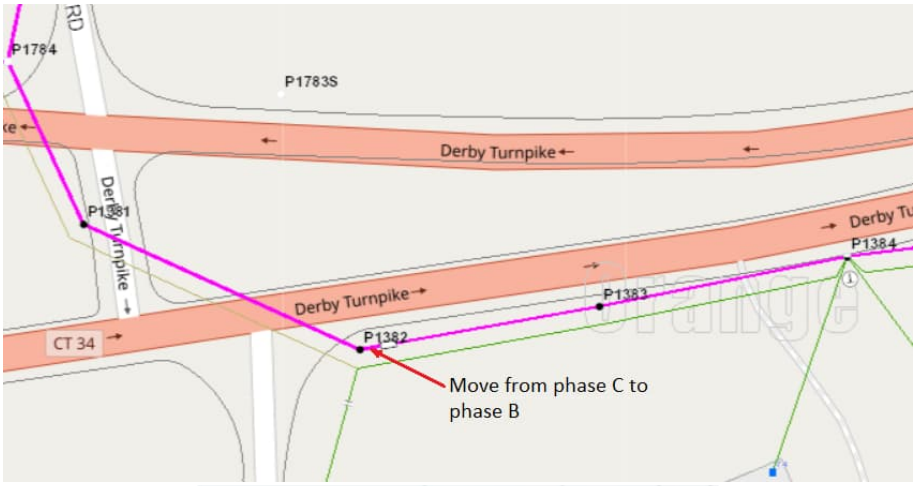




Figure 3: Load balancing at pole #1382 on Derby Turnpike

Option 5

Note this option is not a standalone option to resolve capacity issue. This option recommends extending the existing underground electric duct-line facilities by approximately 350 ft. onto West Spring Street from Campbell Avenue to Vista switch riser conduits to replace circuit 626 aerial cable to the hospital with 3-500 Cu EPR underground cable in splicing chamber and

duct-line system and remove all circuit 626 aerial cable and both underground cable risers. Note there will still be one tie point remaining on circuit 626 with circuit 631. This option is a supplementary option to be combined with options 1, 2, and 4B to help the hospital with meeting or exceeding federal resiliency requirements. This option does not affect the normal or contingency circuit ratings for this area.

Substation Loading Analysis

As mentioned before, the primary circuit for this new load is supplied by Elmwest Substation. The rating for Elmwest Substation is 92 MVA. Due to all the reconfigurations suggested in Option 4B, there will be around 2.5 MW new load transferred from Allings Crossing substation to Elmwest substation. The figure below shows the 10-year forecasted load versus the rating for Elmwest Substation, including the proposed 3.5 MW load addition for VA Hospital. The new customer load is assumed to go online for the 2027 peak. The load forecast depicted in Figure 2 below represents a normalized extreme weather scenario (90/10), which is only expected to occur once every ten years. As can be seen from the figure below, Elmwest Substation has enough capacity to supply this load increase even when considering an extreme weather forecast scenario.

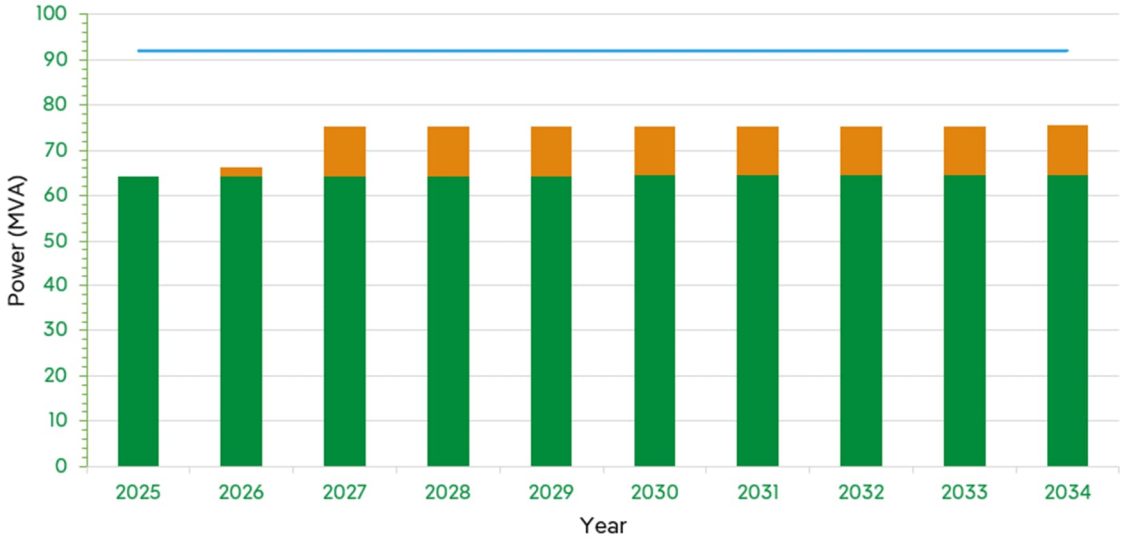


Figure 4: Elmwest Substation 10-Year 90/10 Load versus Rating

Voltage Change Analysis

In this section the voltage profiles of the circuits reconfigured to accommodate the new load are displayed under both initial and final conditions. As seen from Figures 5 to 14, there won't exist any voltage violation after Option 4B is implemented. It is notable that all the voltage values located below the lower threshold bar (117V) are secondary nodes, and thus, the primary voltage bound (117V to 126V) does not apply to them.

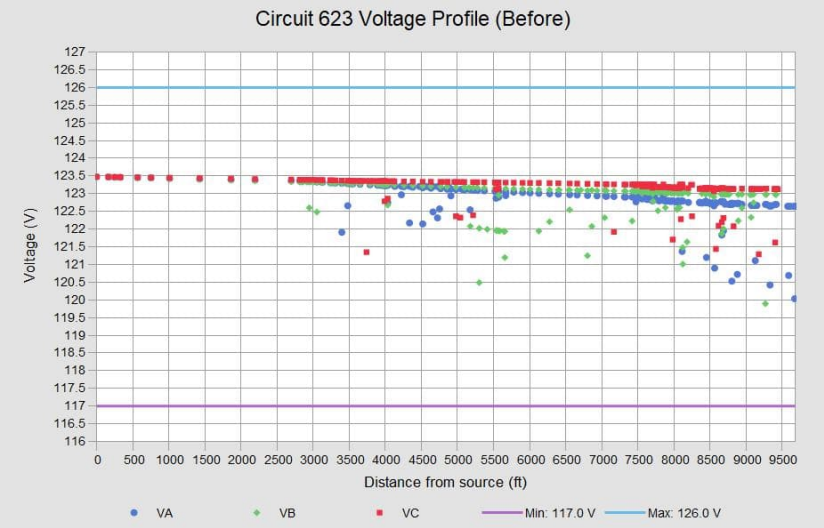


Figure 5: The voltage profile of circuit 623 in existing configuration

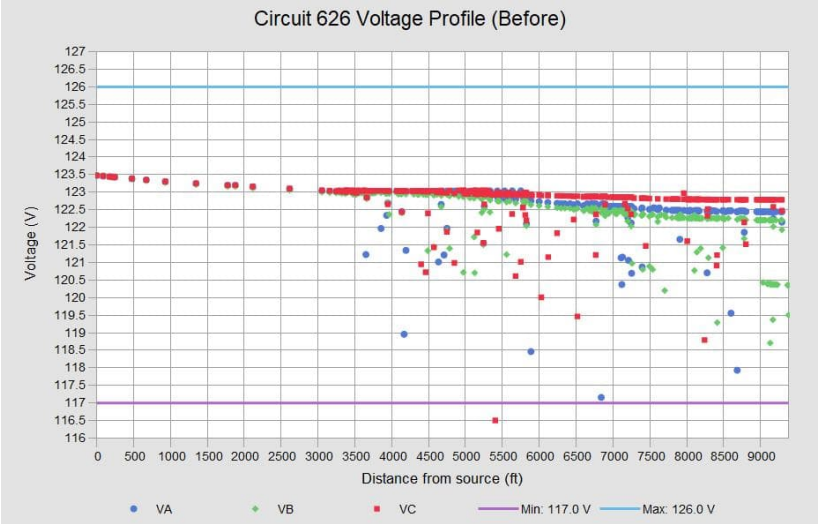


Figure 6: The voltage profile of circuit 626 in existing configuration

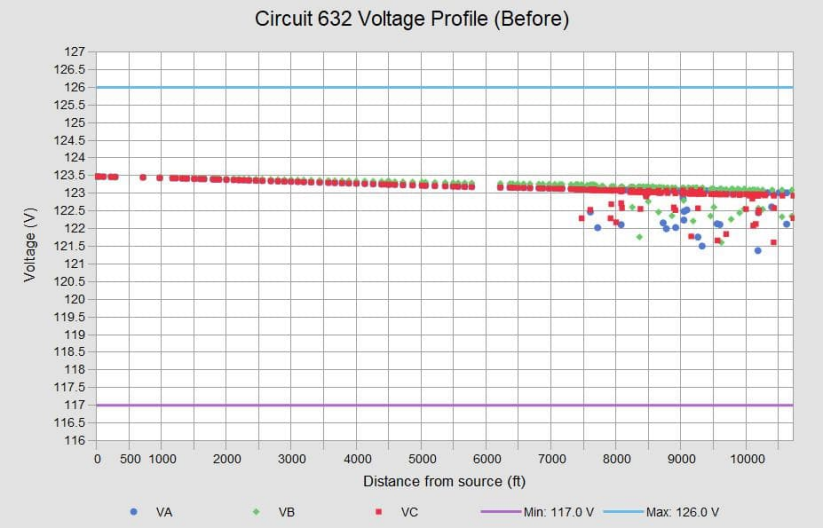


Figure 7: The voltage profile of circuit 632 in existing configuration

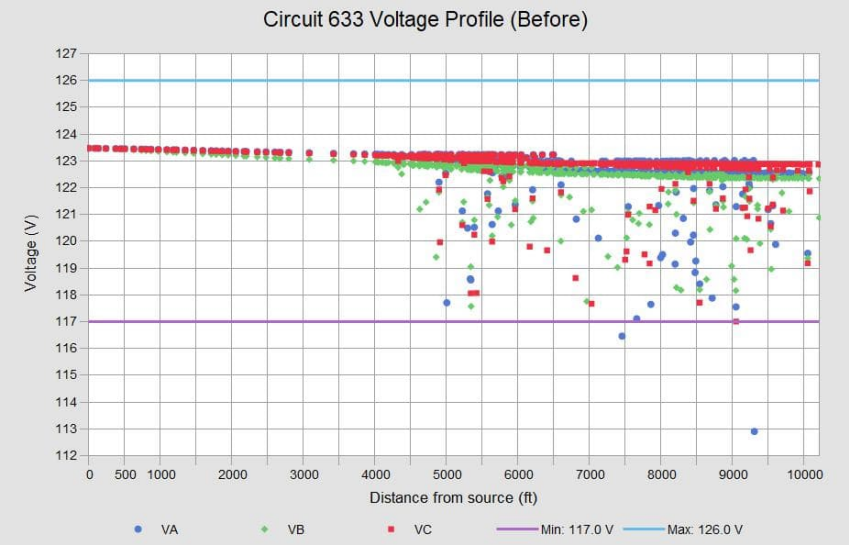


Figure 8: The voltage profile of circuit 633 in existing configuration

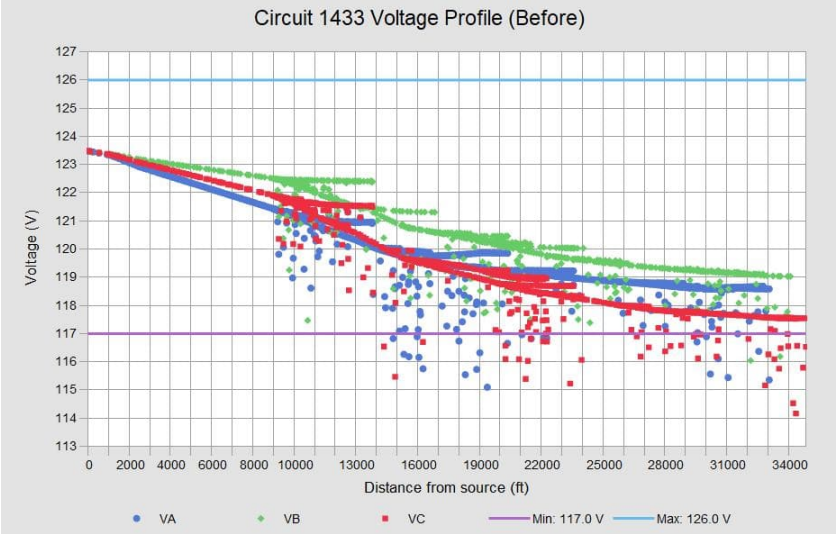


Figure 9: The voltage profile of circuit 1433 in existing configuration

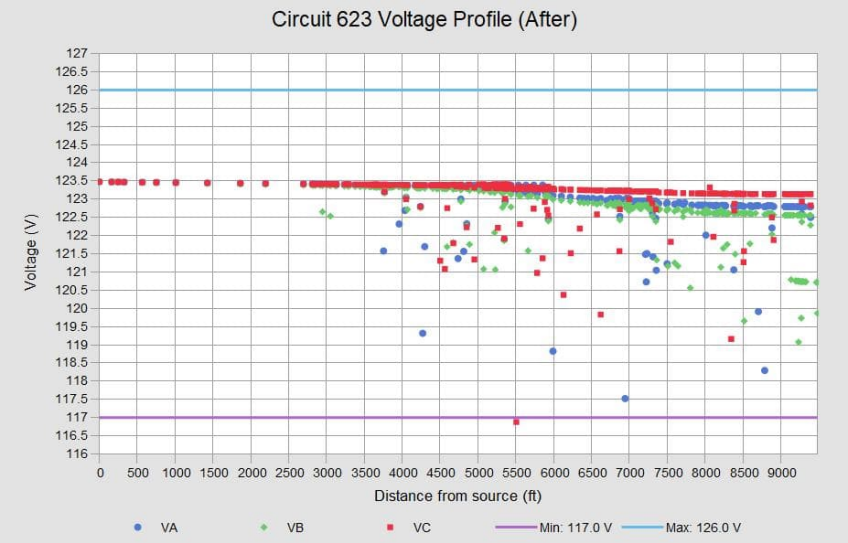


Figure 10: The voltage profile of circuit 623 after Option 4B implemented

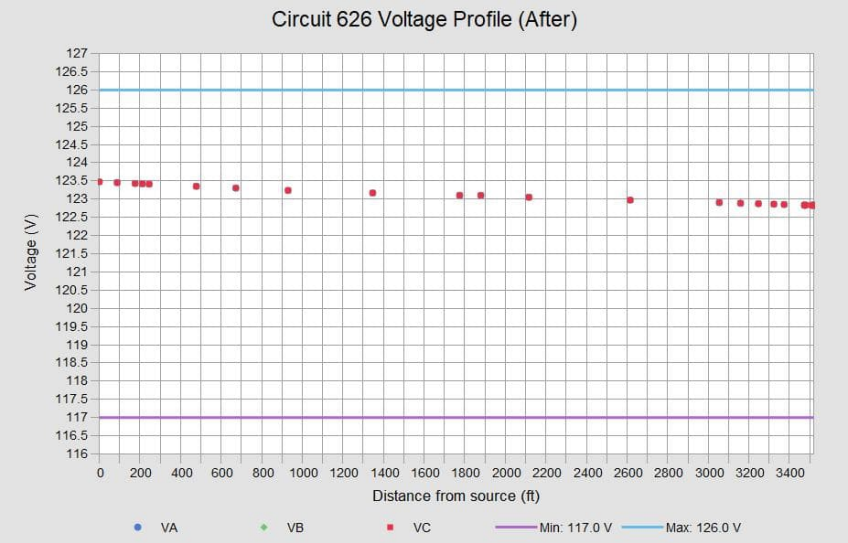


Figure 11: The voltage profile of circuit 626 after Option 4B implemented

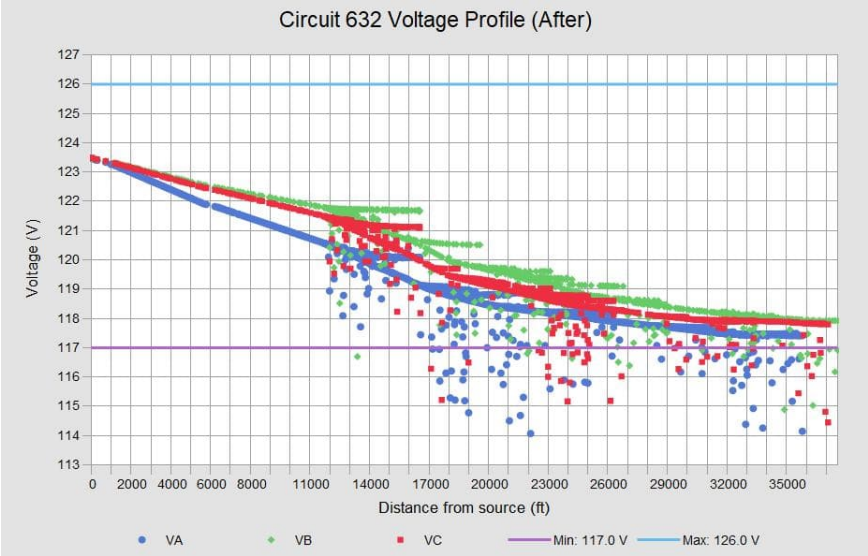


Figure 12: The voltage profile of circuit 632 after Option 4B implemented

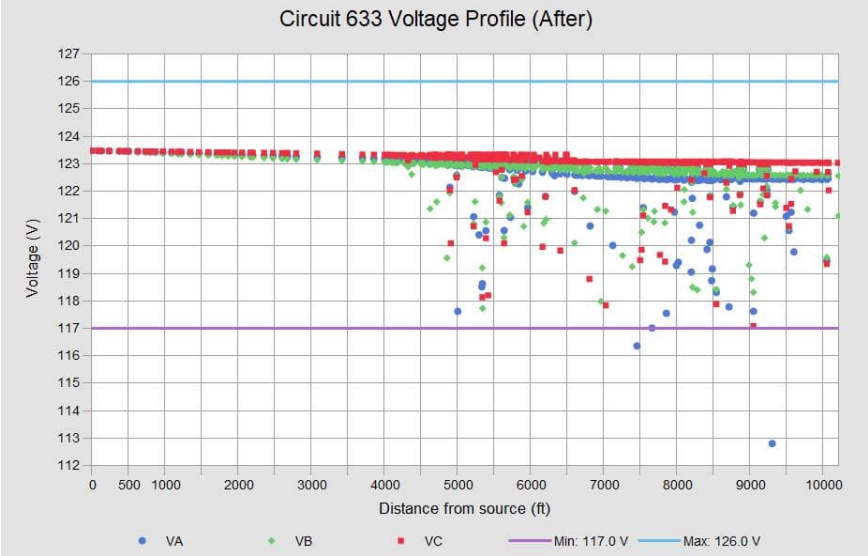


Figure 13: The voltage profile of circuit 633 after Option 4B implemented

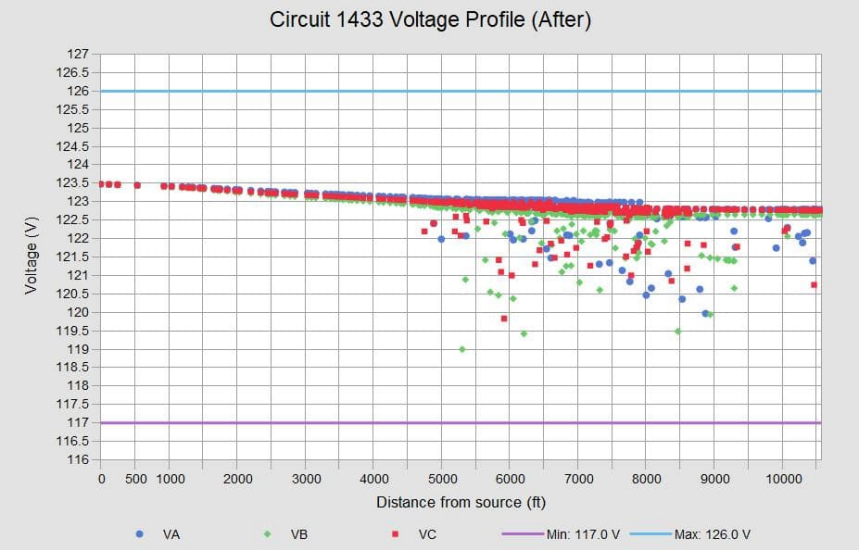


Figure 14: The voltage profile of circuit 1433 after Option 4B implemented

Protection Analysis

The load addition may impact the circuit overcurrent protection settings. Protection & Control's review and verification is required for the existing reclosers, and circuit breakers.

Customer Threshold Analysis

To reduce or maintain SAIFI across the system, circuits shall not have customer counts greater than 2,500. The customer load addition does not include adding additional meters to serve the new load. Also, customer count on all the circuits involved in the project shows that neither of the options will compromise the customer threshold. Table 1 lists the number of customers for circuits that are reconfigured in Option 4B.

Table 8: Customer count for circuits impacted in Option 4B

Circuit	Substation	Existing number of customers	Number of customers after Option 4B executed
623	Elmwest	1030	1143
626	Elmwest	1134	1 (dedicated)
632	Elmwest	410	1911
633	Elmwest	1452	1273
1433	Allings Crossing	1911	1609

Assumptions

Previous year peak loads were used for the loading analysis. The duct-line rating analysis was updated for Elmwest Substation and subsequently the circuit ratings changed. As such, the new ratings were used for this analysis. This study does not include any customer's Distributed Generation (DG) application. Any DG application would require a separate study request.

Conclusion

The analysis shows that the new hospital load addition of 3.5 MW on Elmwest substation is feasible provided the load is supplied via Option 4B so that the new load will not adversely affect the circuit's voltage performance or thermal rating, and the decrease in ratings of other circuits in the same duct bank system is mitigated. Furthermore, the substation will not approach its firm rating of 92 MVA with this new customer load addition. It is noted that Options 3 and 4 have been deemed non-constructable and Options 1 and 2 do not meet the requirement of having the back-up feeder from a different substation. Implementing option 4B and option 5 provides alternate substation supplies for the hospital and therefore is the recommended approach for supplying power to the new load. This study and associated analyses are valid for six (6) months and may need to be reviewed and revised due to, but not limited to, changes in UI system changes, customer load additions, or changes to UI company policies. A summary of the upgrades considered to enable new load addition are in appendix 5. As noted in UI Guidebook of Requirements for Electric Service; when the estimated expenditures of the company to provide service to a customer's premises exceed the estimated income to be derived from that service, additional payments by the customer may be required. The company may require the

customer to pay, in advance, the estimated cost difference to provide such service or to guarantee a minimum annual payment for a term of years as stated in UI's Terms and Conditions (CPUCA No. 647, Item 4).

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Appendix 1: Customer Load Form (New Surgical and Clinical Tower)

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Load Information Form

NOTE: REFER TO THE INSTRUCTIONS ON PAGE TWO OF THIS FORM. FAILURE TO FOLLOW THESE INSTRUCTIONS OR COMPLETE ALL NECESSARY INFORMATION MAY RESULT IN DELAYS.

PROJECT NAME: New Surgical and Clinical Tower
OWNERS NAME: US Dept. of Veterans Affairs
ADDRESS: W. Spring St. 1, West Haven, CT 06516
ADDRESS: W. Spring St. 1, West Haven, CT 06516
NEAREST INTERSECTION: W. Spring St. and Campbell Ave.
PRINCIPAL CONTACT: ???
BUSINESS TYPE: Federal Healthcare
TELEPHONE: 206-932-5711

CONNECTED LOAD INFORMATION

Load Type	Connected Loads (KW) 1 Phase	Connected Loads (KW) 3 Phase
Lighting	324	
Water Heating		
Electric Heat		
Air Conditioning		810
Other HVAC Equip.		included in AC
Refrigeration		
Cooking		60
Receptacles	729	
Motor Load (HP)		306
Process Heat		
Other (Describe)		
- Misc./Medical Equip.		843
- Future Growth	265	500
Totals	1,318	2,519

SERVICE INFORMATION
Overhead ☐ Underground ☒
Service Size EXISTING Amps
Voltage Desired EXISTING - 13.8kV
3 Phase 4 Wire

SECONDARY CONDUCTORS
Conductor Size
Number of Sets
Copper ☐ Aluminum ☐

METERING INFORMATION
Total # of Meters EXISTING
Total # of Customers EXISTING
Secondary Meter Desired ☐
Primary Meter Desired ☐

BUILDING STRUCTURE
☐ New ☐ Renovation
☒ Add ☐ Existing
Square Feet 162,000 + 32,400 future
Heating Fuel Natural gas boiler with fuel oil backup

TRANSFORMERS DESIRED
☐ Padmount
☐ Pole Mount
☐ Transclosure
☐ Vault
NO NEW TRANSFORMATION REQUIRED

METER PROVISIONS

Quantity	Size	Phase	Wire

SCHEDULING INFORMATION
Construction Start Date:
Date Service Required:
Building Occupancy Date:

LARGEST MOTORS

	1 Phase	3 Phase
HP		
LRA		
Quan.		

REMARKS:

Design is in conceptual infancy. The building footprint is not yet set and no equipment is selected. Loads are preliminary and have been established based on engineering practice and square footage allowances. This load information will become refined as the design progresses and new load information will be provided to the utility as resolution is established. Chilled water and steam will be generated at a campus level and not at the new tower.

SUBMITTING FIRM: IMEG Corp. **DATE:** 10/12/2022
ADDRESS: 672 E. Royalton Rd. Cleveland, OH. 44147
PROJECT HEAD: Christopher Campbell, P.E. **TELEPHONE NO.:** 440-546-9776

Note: Contact Customer Engineering to obtain delivery lead times for equipment such as transformers, switches and primary cable.

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Appendix 1: Customer Load Form Cont'd. (Boiler and Chiller Upgrades)

United Illuminating

Load Information Form

NOTE: REFER TO THE INSTRUCTIONS ON PAGE TWO OF THIS FORM. FAILURE TO FOLLOW THESE INSTRUCTIONS OR COMPLETE ALL NECESSARY INFORMATION MAY RESULT IN DELAYS.

PROJECT NAME: Hospital Boiler & Chiller Upgrades
OWNERS NAME: US Dept. of Veterans Affairs
ADDRESS: 1 W Spring St, West Haven, CT 06516
ADDRESS: 1 W Spring St, West Haven, CT 06516
NEAREST INTERSECTION: W. Spring St and Campbell Ave.
PRINCIPAL CONTACT:
BUSINESS TYPE: Federal Healthcare
TELEPHONE: 206-932-5711

CONNECTED LOAD INFORMATION

Load Type	Connected Loads (KW) 1 Phase	Connected Loads (KW) 3 Phase
Lighting	23	
Water Heating		15
Electric Heat		
Air Conditioning		3096
Other HVAC Equip.		20
Refrigeration		
Cooking		
Receptacles	90	
Motor Load (HP)		1219
Process Heat		
Other (Describe)		
Totals	113	4350

SERVICE INFORMATION
Overhead ☐ Underground ☐
Service Size Amps
Voltage Desired
Phase Wire

SECONDARY CONDUCTORS
Conductor Size
Number of Sets
Copper ☐ Aluminum ☐

METERING INFORMATION
Total # of Meters
Total # of Customers
Secondary Meter Desired ☐
Primary Meter Desired ☐

BUILDING STRUCTURE
☒ New ☐ Renovation
☐ Add ☐ Existing
Square Feet
Heating Fuel

TRANSFORMERS DESIRED
☐ Padmount
☐ Pole Mount
☐ Transclosure
☐ Vault

METER PROVISIONS

Quantity	Size	Phase	Wire

SCHEDULING INFORMATION
Construction Start Date:
Date Service Required:
Building Occupancy Date:

LARGEST MOTORS

	1 Phase	3 Phase
HP		200
LRA		2430
Quan.		3

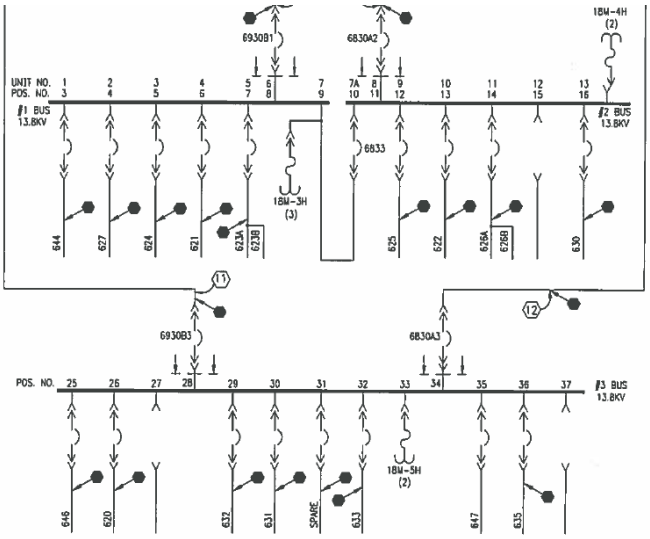
REMARKS:

SUBMITTING FIRM: 3 DATE:
ADDRESS:
PROJECT HEAD: TELEPHONE NO.:

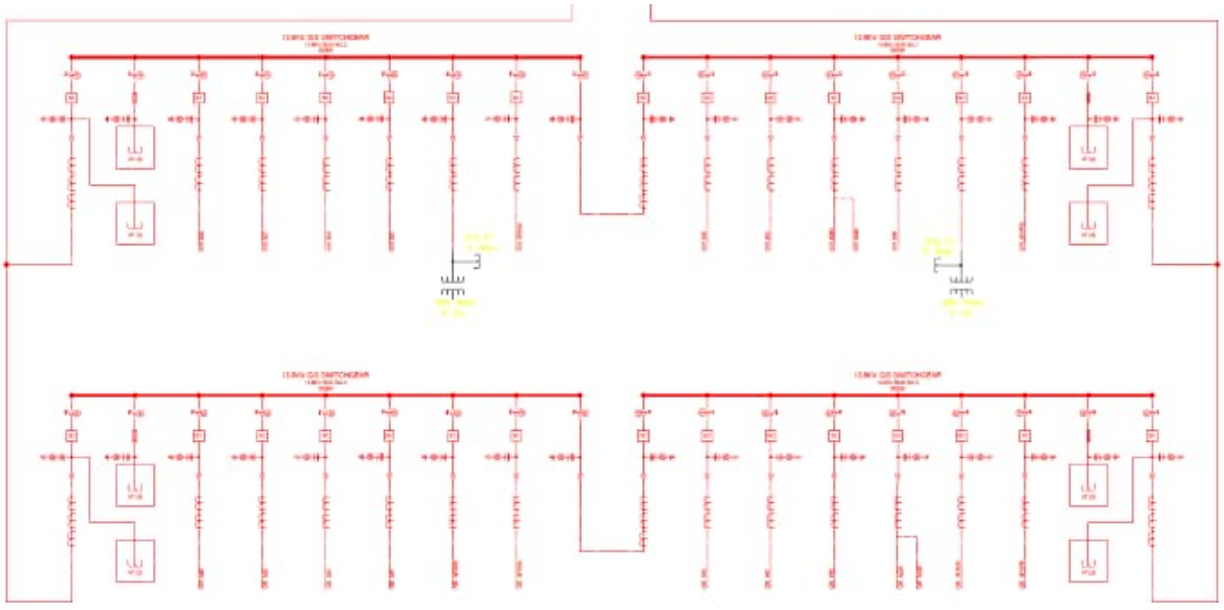
Note: Contact Customer Engineering to obtain delivery lead times for equipment such as transformers, switches and primary cable.

Appendix 2: Elmwest Substation One-line

Existing

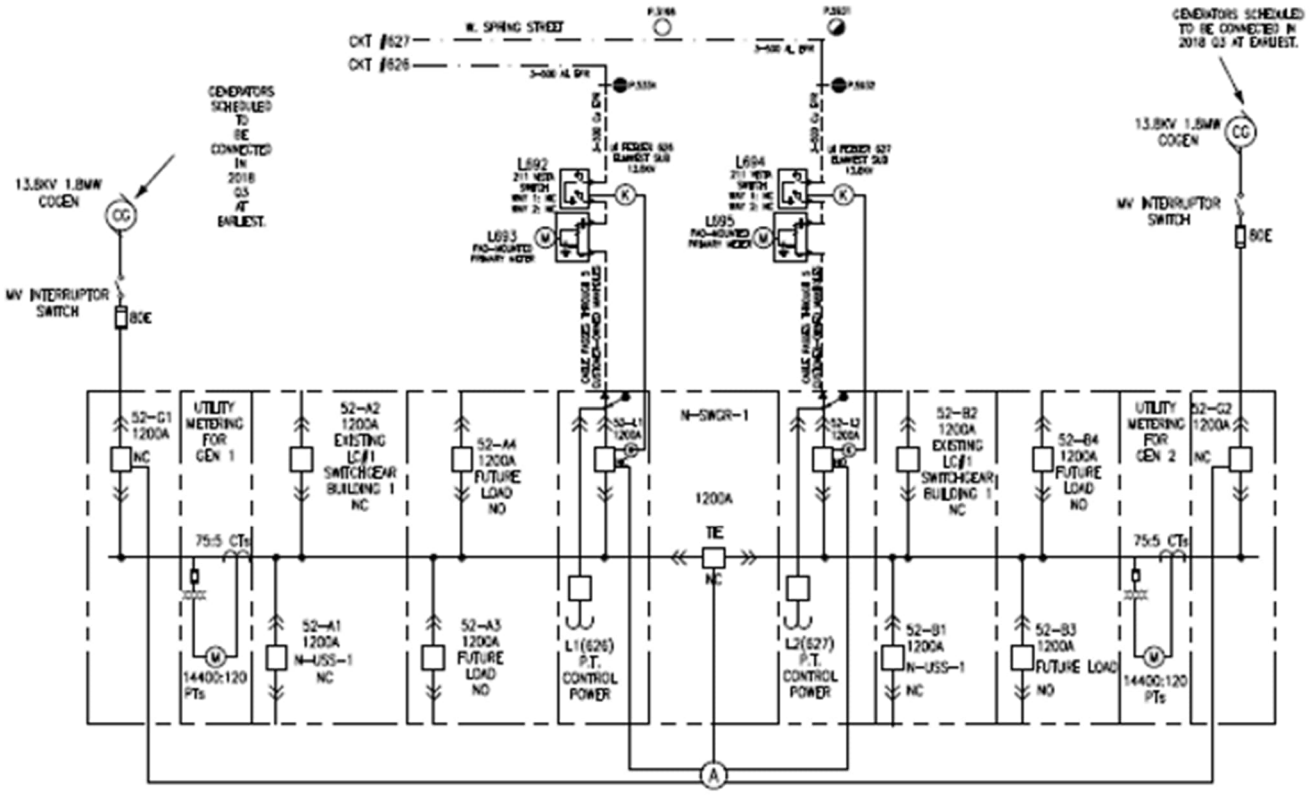


Proposed (Preliminary design for split bus project)

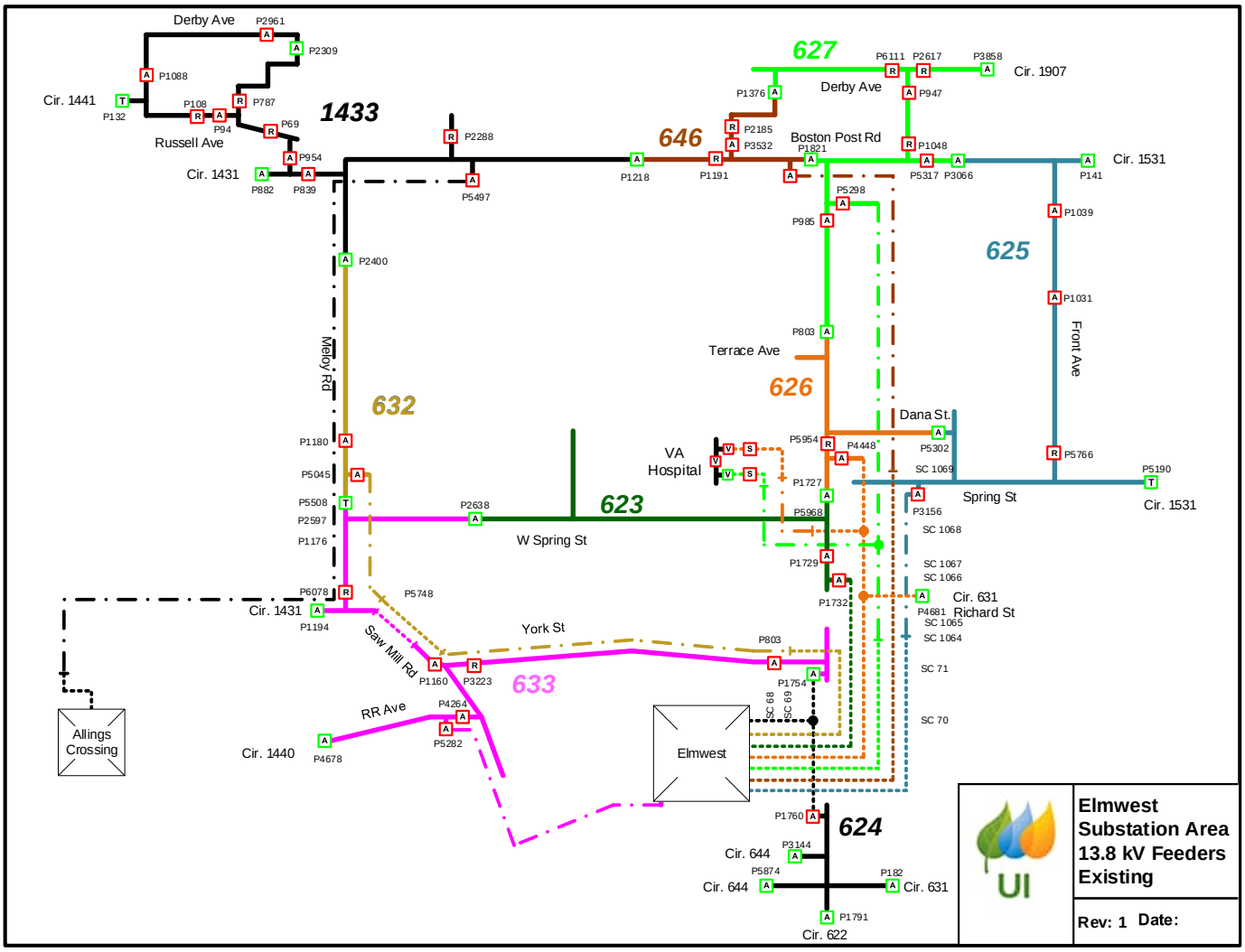


VA Hospital, West Haven New Customer Load Report

Appendix 3: VA Hospital Customer One-line

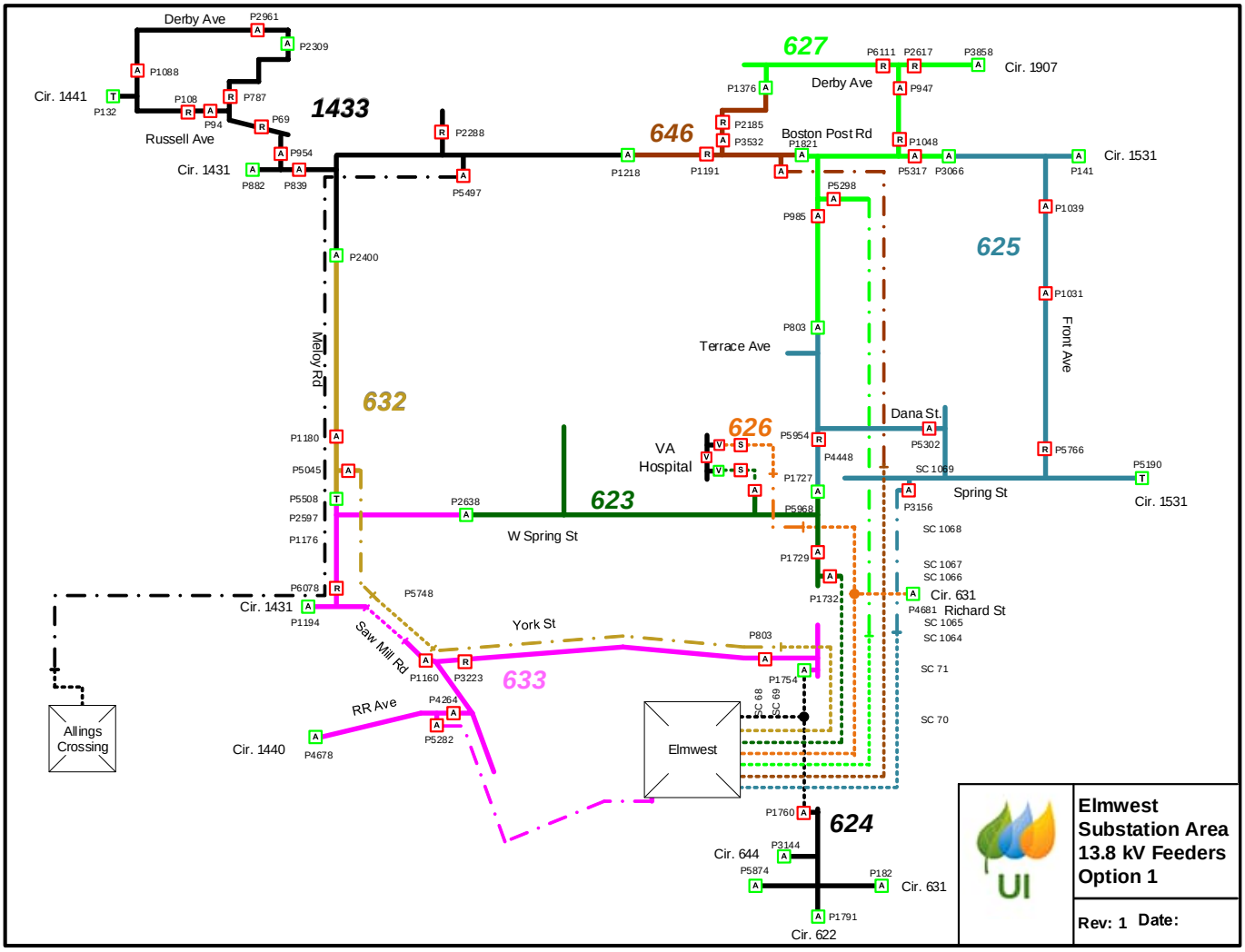


Appendix 4: Elmwest Substation Area One-lines for VA Hospital Study



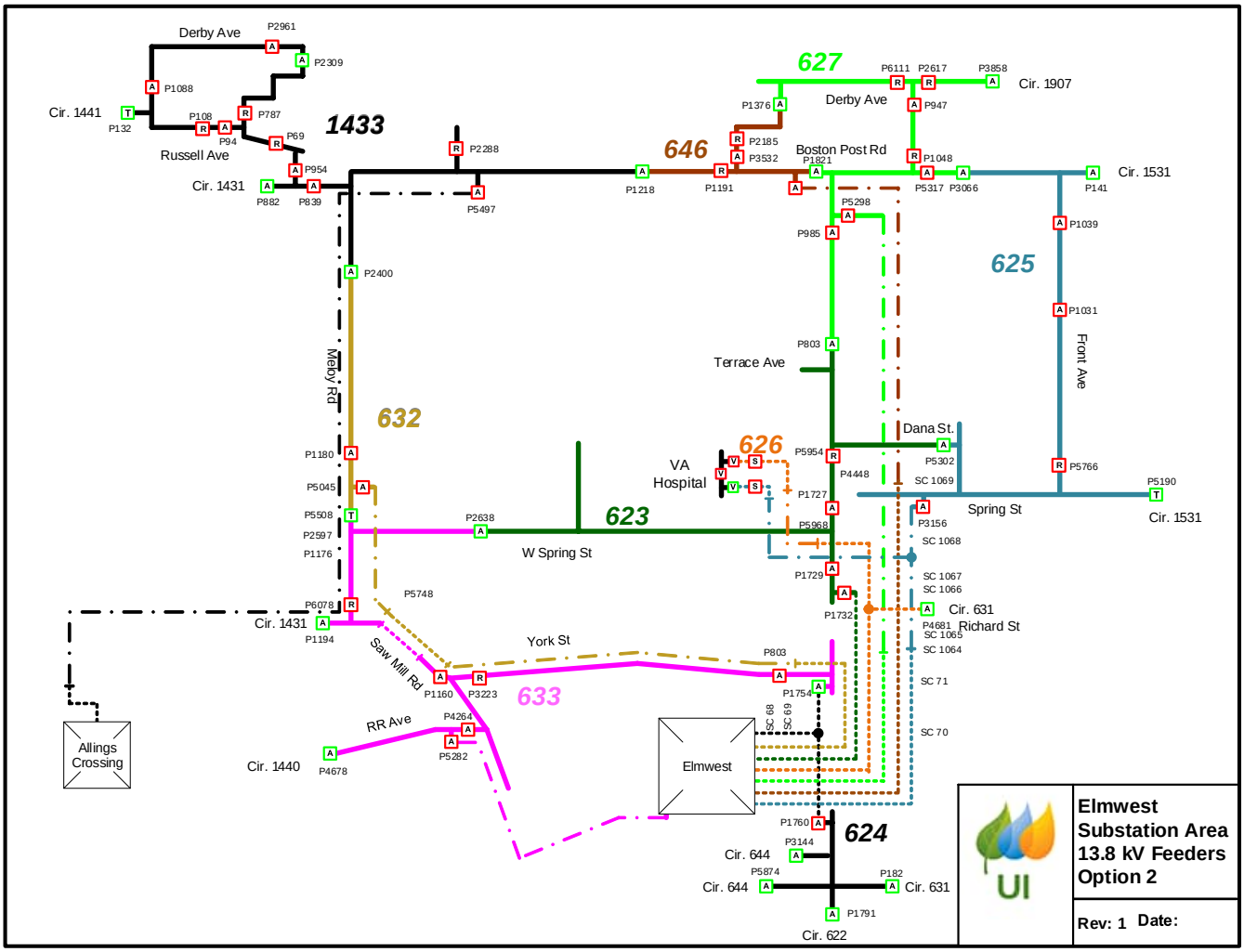
VA Hospital, West Haven New Customer Load Report

Appendix 4: Elmwest Substation Area One lines for VA Hospital Study Cont'd.



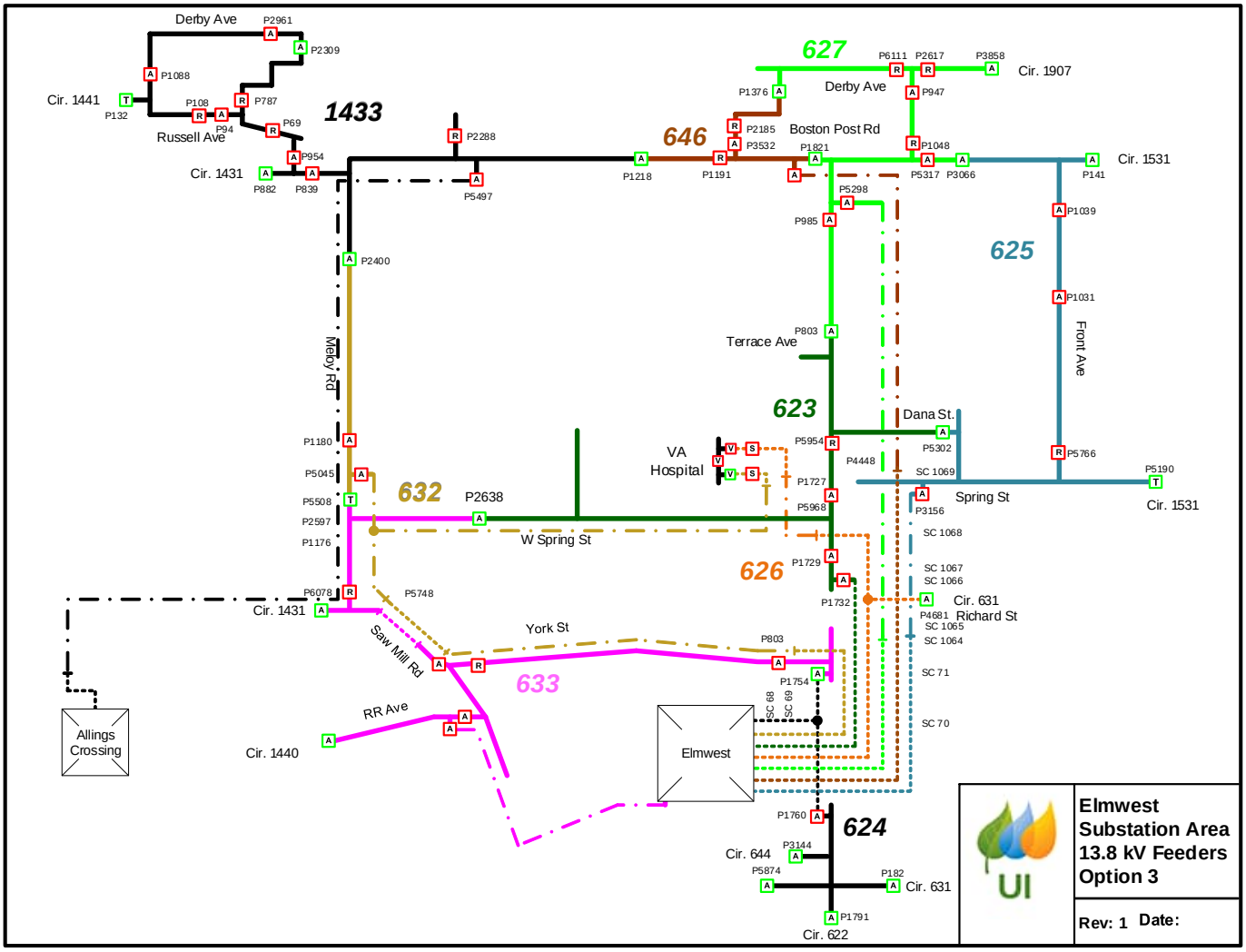
VA Hospital, West Haven New Customer Load Report

Appendix 4: Elmwest Substation Area One lines for VA Hospital Study Cont'd.



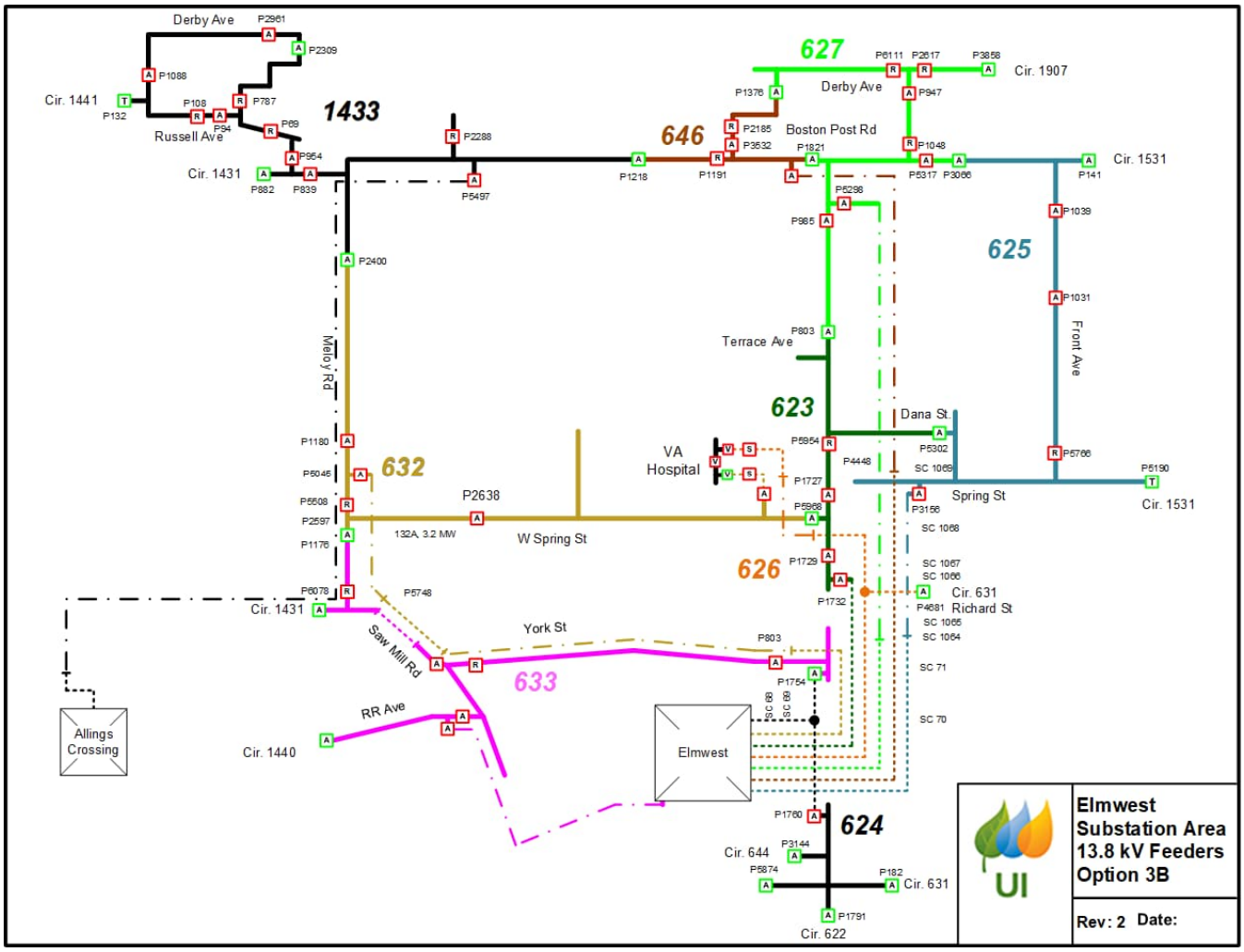
VA Hospital, West Haven New Customer Load Report

Appendix 4: Elmwest Substation Area One lines for VA Hospital Study Cont'd.



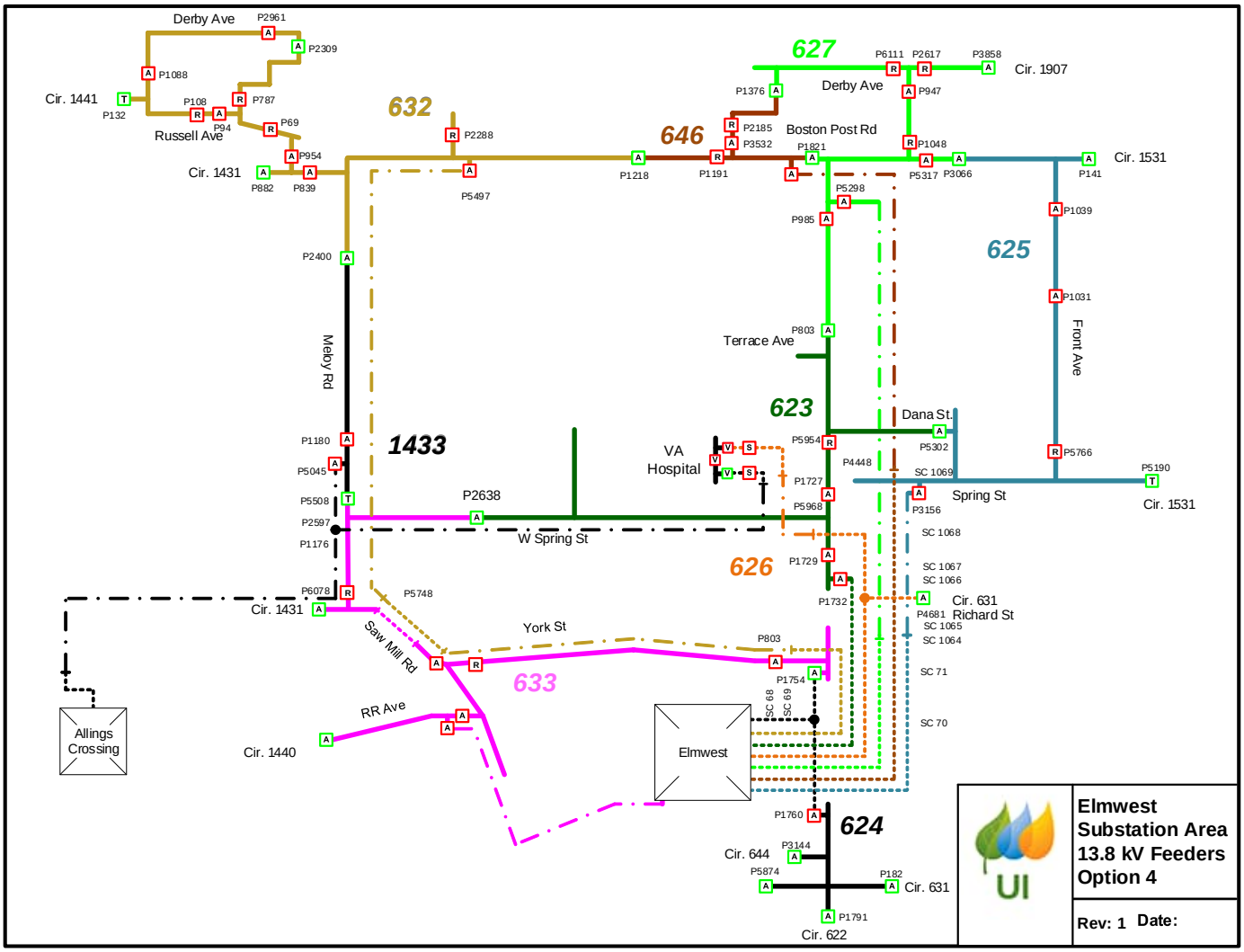
VA Hospital, West Haven New Customer Load Report

Appendix 4: Elmwest Substation Area One lines for VA Hospital Study Cont'd.



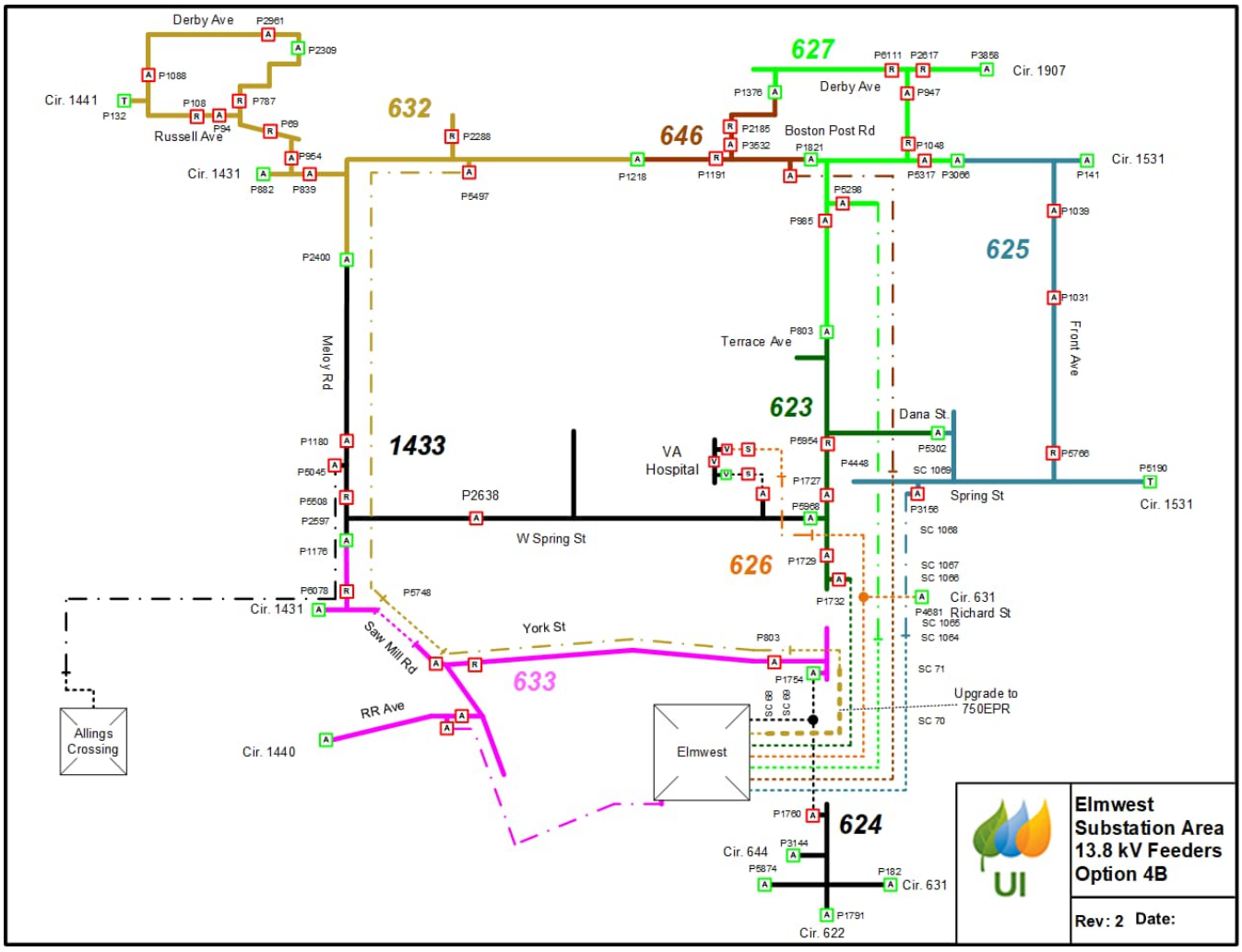
VA Hospital, West Haven New Customer Load Report

Appendix 4: Elmwest Substation Area One lines for VA Hospital Study Cont'd.



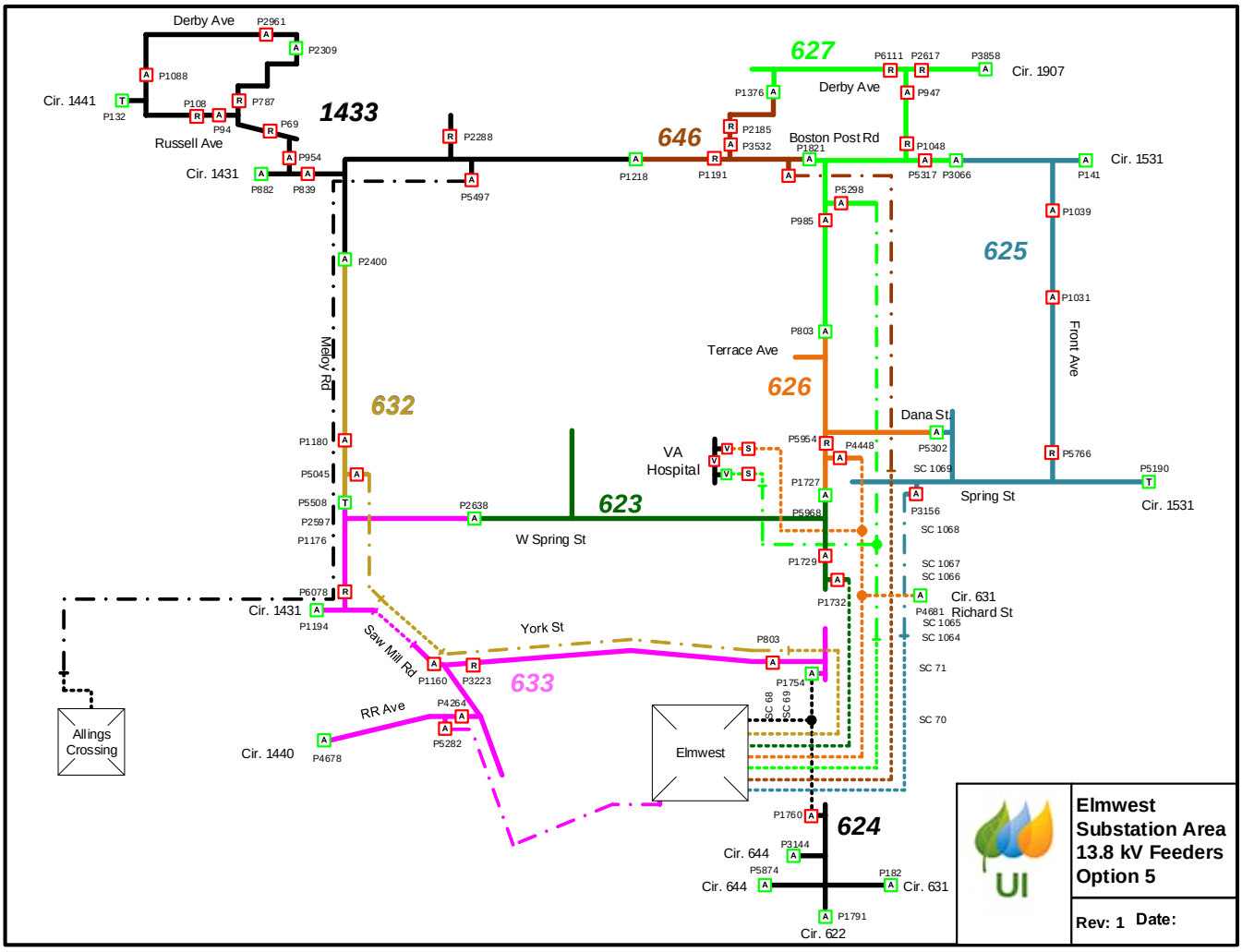
VA Hospital, West Haven New Customer Load Report

Appendix 4: Elmwest Substation Area One lines for VA Hospital Study Cont'd.



VA Hospital, West Haven New Customer Load Report

Appendix 4: Elmwest Substation Area One lines for VA Hospital Study Cont'd.



VA Hospital, West Haven New Customer Load Report

Appendix 5: Upgrade Summary

VA Hospital New Load Option Summary		Different Substation Sources	Ckts < 50 ft. apart (ft)			Upgrade Cost	Customer Cost	Buildable
Option	Description		OH	**UG**	OH + UG	($ K)	($ K)	
	Adding new load to existing Ckt configuration without any upgrades	NO	2,375	610	2,985	NA	NA	NO
1	Offload Ckt 626 to Ckt 625 & make Ckt 623 alternative source via O.W.	NO	275	2,297	2,572	214.3	53.1	YES
2	Offload Ckt 626 to Ckt 623 & make Ckt 625 alternative source via A. Cable	NO	275	610	885	176.7	15.5	YES
3	Offload Ckt 626 to Ckt 623 & make Ckt 632 alternative source via A. Cable	NO	0	610	610	879.4	718.2	NO
3B	Offload Ckt 626 to Ckt 623 & make Ckt 632 alternative source via O.W.	NO	275	610	885	161.2	0 *	YES
4	Offload Ckt 626 to Ckt 623 & make Ckt 1433 alternative source via A. Cable	YES	0	0	0	839.6	678.4	NO
4B	Offload Ckt 626 to Ckt 623 & make Ckt 1433 alternative source via O.W.	YES	275	0	275	417.2	256	YES
5	Underground Ckt 626 aerial cable	NO	0	NA	NA	914.8	914.8	YES
4B+5						1,332	1,170.8 *	

275' values marked in red drop to zero if Option 5 is completed

Assuming upgrades are completed in 2027, * - customer cost subject to minimum bill calculation for service upgrade

Upgrade costs include overheads, CIAC, & 25 % contingency, but do not include environmental remediation or ledge removal costs